NS-3 NB-IoT Simulator User Guide

Introduction

This guide provides students with a fully prepared environment for running NS-3 NB-IoT simulations using Docker. The goal is to ensure a consistent, reproducible setup across all lab machines and student laptops.

The NS-3 simulator is packaged inside a Docker container, eliminating the need for manual installation of dependencies or compilation on your local system.

1. Overview

The NS-3 NB-IoT simulator is provided as a Docker-based environment. Docker ensures that every student runs the same version of NS-3 with identical dependencies. You will:

- Install Docker
- Build the NS-3 simulator container
- Run the container
- Execute NB-IoT simulations

2. Installing Docker

Docker allows applications to run inside isolated containers. Follow the official Docker installation guide:

https://docs.docker.com/engine/install/

If you prefer you can install its GUI (https://docs.docker.com/desktop/). In either case, choose your operating system (Windows, macOS, Linux) and follow the steps provided.

Verify Installation

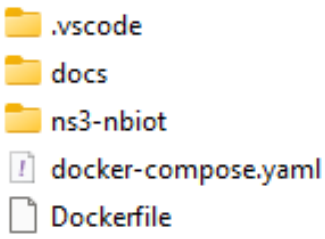
Open a terminal and run:

```
docker --version
docker compose version
```

You should see version numbers printed.

3. Project Structure

Your instructor will provide a folder[1] containing:



These files define the NS-3 simulator environment.

You do not need to modify them.

4. Building the NS-3 Simulator Container

Open a terminal and navigate to the project folder:

```
cd <project-folder>
```

Build the container:

```
docker compose build
```

This step:

- Downloads Ubuntu 22.04
- Installs NS-3 dependencies
- Clones the NB-IoT NS-3 repository
- Builds NS-3 using waf

The *build* may take several minutes, and your computer needs to be connected to the Internet.

[1] You can also download it using **git clone https://github.com/h3dema/ns3-simulator.git**

5. Running the Container

Start an interactive session inside the NS-3 environment:

```
docker compose run ns3
```

You will be placed inside the workdir defined in docker-compose.yaml. By default, it is **/opt/ns3-nbiot**. This directory contains the full NS-3 source tree.

There is **another option** to run the container and **automatically remove it from the disk when you exit it**. Use it with care, because the container is destroyed with no leftovers (you lose anything you left inside the container's disk), only the original image is intact.

```
docker compose run --rm ns3
```

6. Running NS-3 Simulations

Inside the container, simulations are executed using waf:

```
./waf --run <program-name>
```

For example, to run a simulation called "my-simulation.cc"[2] in the scratch folder:

```
./waf --run scratch/my-simulation
```

[2] This is only an example. There is no file with this name in the repository.

7. Exiting and Re-entering the Container

To exit the NS-3 environment:

```
exit
```

To re-enter later:

```
docker compose run ns3
```

8. Optional Cleanup

Remove the built image:

```
docker image rm ns3-simulator-ns3
```

Remove stopped containers.

```
docker container prune
```

If you run the container with the --rm option, every time you “exit” the container, it is automatically removed (and you cannot reenter it).

9. NetAnim

NetAnim is an offline animation tool built with the Qt framework. Rather than running a live simulation, it takes an **XML trace file** generated from a previously executed simulation and visually animates the network topology and traffic flow. In this guide, we are assuming that we are going to install NetAnim in the host OS (The examples below are on an Ubuntu host).

Prerequisites & Installation

Before building NetAnim, install its required build tools and Qt5 development packages. All the commands in this section were tested in an Ubuntu 22 LTS desktop and in a WSL2 Ubuntu environment. More information can be found in the documentation.

Step 1: Install Dependencies

Open your Linux terminal and execute:

```
sudo apt update
sudo apt install -y build-essential qtbase5-dev qtchooser qt5-qmake qtbase5-dev-tools cmake git mercurial
```

Step 2: Download NetAnim

Clone the official NetAnim source repository using Mercurial (hg):

```
hg clone http://code.nsnam.org/netanim
```

Step 3: Build NetAnim

NetAnim uses qmake (Qt's build system) for compilation and requires **Qt version 5.4 or higher**. Enter the cloned directory. Clean previous builds to ensure no stale object files remain. If it is your first compilation, just skip this step. Generate the Makefile and nuild the source code into an executable.

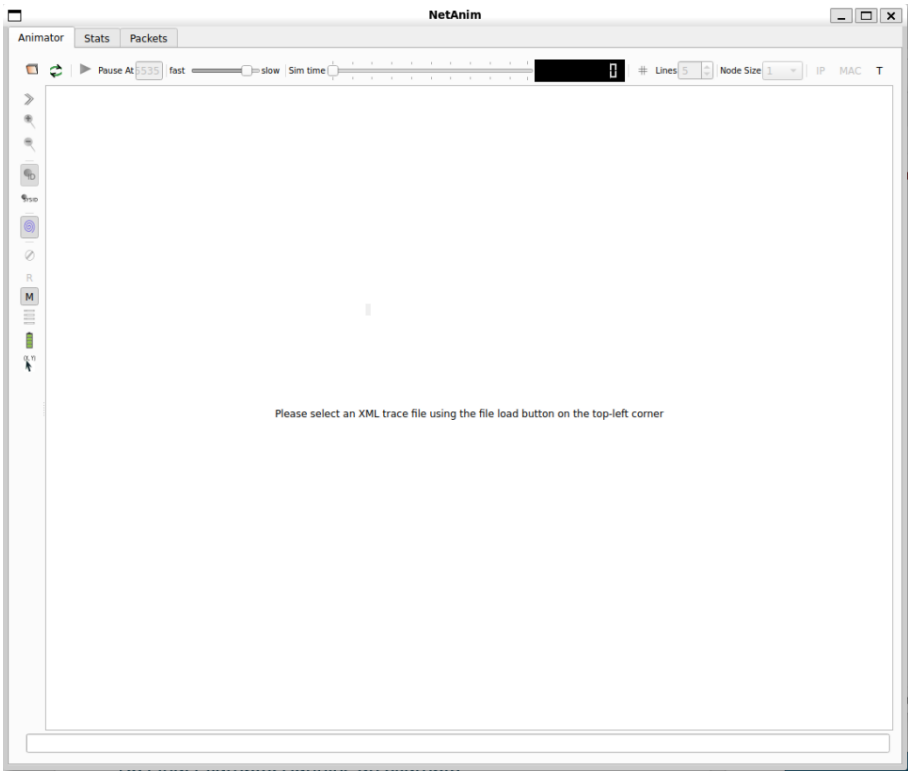
```
cd netanim
make clean
qmake NetAnim.pro
make
```

Note on Compilation Warnings: You may see several compiler warnings during the make step. **This is normal.** As long as the build finishes without an Error, your *./NetAnim* executable is ready.

Step 4. Launch the GUI:

Execute the binary from your terminal and it will show the interface below.

```
./NetAnim
```



1. Load the File in NetAnim

1. Launch NetAnim using ./NetAnim.
2. Click the **Folder Icon** (top-left toolbar) to open the file browser.
3. Select and open your generated **animation.xml** file.

2. Play the Animation

- Click the **Play button** in the top toolbar to start playback. (Button turns green when an animation is loaded).
- Use the **Speed slider** to adjust the simulation rate or pause at specific timestamps to analyze packet transmissions.

10. Running an example

In this lab, you will explore energy consumption and state transitions in Cellular IoT (5G mMTC) devices using the ns-3 network simulator. You will run a simulation scenario where an IoT device equipped with an energy harvester sends small data packets over a cellular network to a remote server.

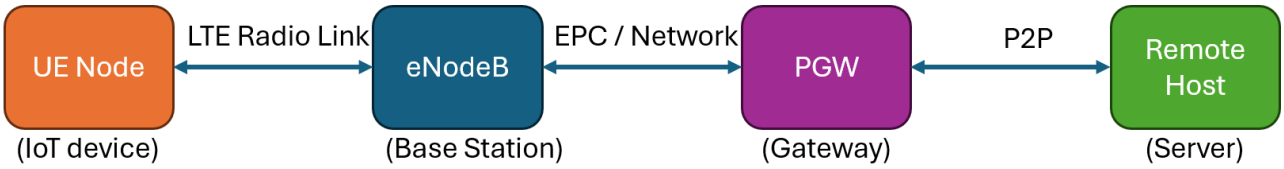
Lab Objectives

By the end of this lab, you will be able to:

- Understand the architecture of an LTE/5G Cellular IoT simulation in ns-3.
- Run customized network simulations inside a Docker container.
- Observe state changes (ACTIVE vs. INACTIVE) driven by the energy model.
- Locate and inspect generated log files and packet traces.

Network Scenario Architecture

The simulation (nb-scenario5.cc) models a typical IoT deployment:



1. **User Equipment (UE):** A static IoT device placed in a cell with a radius of up to 2500m. It operates on an **Energy Markov** battery model starting at 3.3 V.
2. **eNodeB (Base Station):** Fixed at the center of the cell to handle cellular coverage.
3. **Core Network & Remote Host:** Packets are routed through the LTE Evolved Packet Core (EPC) to a remote host over a high-speed link (100 Gbps).
4. **Traffic Profile:** The IoT device sends 49-byte UDP packets (simulating a 32-byte mMTC payload + CoAP/DTLS headers) every 1 minute.

Ensure your Docker container environment is running. To prevent long simulation times during class, limit the simulation time to **60 seconds** using the --simTime parameter. By default, this parameter is set to 180 minutes. Execute the following command inside your container shell:

```
cd /opt /ns3-nbiot
./waf --run "scratch/nb-scenario5.cc --simTime=60"
```

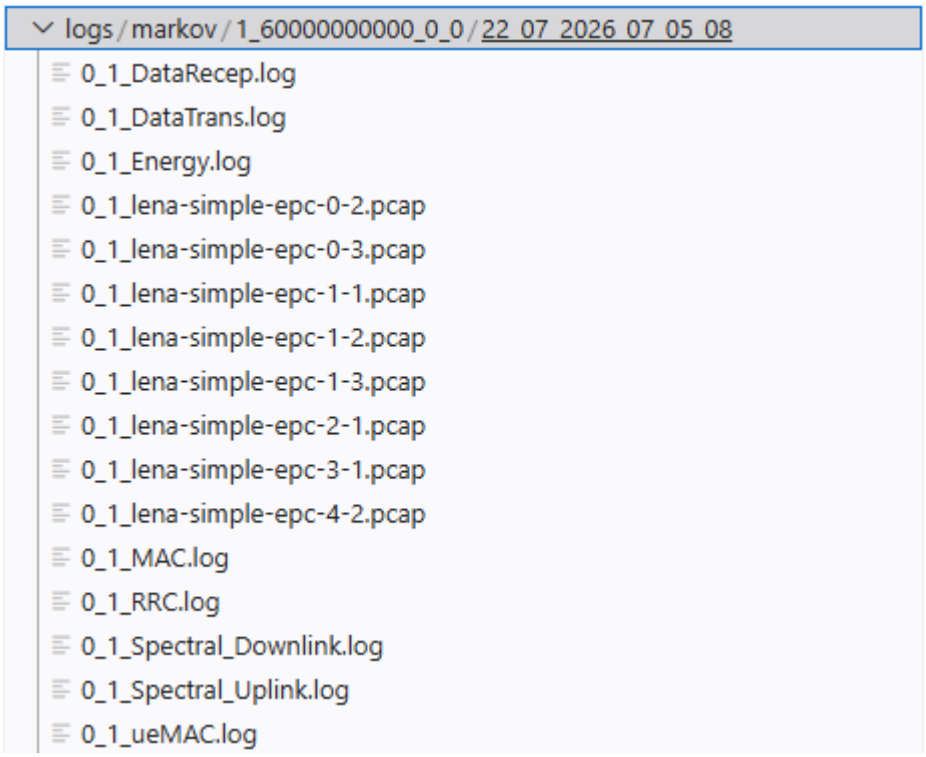
We can define the number of Ues (i.e., IoT devices in the simulation) using the parameter –num_ues. This parameter is, by default, equal to 1.

Inspecting Simulation Outputs & Logs

Once the simulation finishes, output files and traces are automatically organized into structured log directories. Inside the container, all log files are written to “logs” folder inside the ns3 directory.

This container folder is mapped to ./logs on your local host system so you can easily view files using your local text editor or tools.

The folder path automatically structures outputs based on your run configuration:



Log & Trace Output

1. **Console Output:** You will see state transition events printed directly to standard output as the energy levels shift:

```
Waf: Entering directory `/opt/ns3-nbiot/build'
Waf: Leaving directory `/opt/ns3-nbiot/build'
Build commands will be stored in build/compile_commands.json
'build' finished successfully (4.652s)
State ACTIVE   : 1
State INACTIVE: 0
Remote host address: 1.0.0.2
PGW address:
 * Interface #0: 127.0.0.1
 * Interface #1: 7.0.0.1
 * Interface #2: 14.0.0.5
 * Interface #3: 1.0.0.1
Node#0 Position:1394.71,573.577,1.5
EnergyMarkov(0): State machine initialized to 1 with delta0 = 0.7, delta1 = 0.1 at 0 s
EnergyMarkov(0): State machine initialized to 1 with delta0 = 0.7, delta1 = 0.1 at 0 s
EnergyMarkov(0): State machine initialized to 1 with delta0 = 0.7, delta1 = 0.1 at 0 s
Started computation at Wed Jul 22 07:25:48 2026
Number of UEs: 1
EnergyMarkov(5): Updating remaining energy at 0 s
EnergyMarkov(5): State remained as 1 at 0 s
EnergyMarkov(5): Updating remaining energy at 0.01 s
EnergyMarkov(5): State remained as 1 at 0.01 s
EnergyMarkov(5): Updating remaining energy at 0.02 s
EnergyMarkov(5): State remained as 1 at 0.02 s
EnergyMarkov(5): Updating remaining energy at 0.02 s
EnergyMarkov(5): State remained as 1 at 0.02 s
EnergyMarkov(5): Updating remaining energy at 0.02 s
EnergyMarkov(5): State remained as 1 at 0.02 s
EnergyMarkov(5): Updating remaining energy at 0.03 s
```

2. **PCAP Traces:** File captures (such as lena-simple-epc-*.pcap) are stored in the log directory. You can open these with **Wireshark** on your host machine under ./logs/ to analyze packet flows.

Task 1: Knowing the scenario

1. Run the simulation for 30 minutes with 2 IoTs with default settings. Note how many state transitions occur in the console output.
2. Check the log folder under "./logs" on your local machine to verify that your new run generated a timestamped folder containing PCAP files.
3. Open the pcap file using Wireshark[3]. Can you identify the transmitted packets?
4. Open the animation.xml file with NetAnim. Run the animation.

[3] Wireshark is installed inside the container. You can access it calling `wireshark` in the container's ssh shell. More information about Wireshark can be found in the documentation site.

5. Open the “Energy.log”. Plot energy vs. Time[4]. Analyze your findings.

Task 2: Impact of Cell Size (Coverage & Distance)

- **Goal**: Measure how expanding the cell boundary alters energy consumption and system state changes.
- **Execution**: Keep $UE = 1$ and vary --coverage:

```
# Small Cell (500 meters)
./waf --run "nb-scenario5 --simTime=3600 --num_ues=1 --coverage=500"

# Medium Cell (2500 meters - Default)
./waf --run "nb-scenario5 --simTime=3600  --num_ues=1 --coverage=2500"

# Large Cell (5000 meters)
./waf --run "nb-scenario5 --simTime=3600  --num_ues=1 --coverage=5000"

# Rural area (10000 meters from BS)
./waf --run "nb-scenario5 --simTime=3600  --num_ues=1 --coverage=20000"
```

- **Discussion**:
    - Plot the state transition graphs across the 4 setups.
    - *Discussion Point*: How does distance impact energy consumption during active transmission states?

Task 3: Propagation Model Customization (No Recompilation)

- **Goal**: Override propagation parameters using ns3::ConfigStore command-line flags.
- **Execution**: Pass default attribute modifications dynamically:

[4] Check the script `py-codes/plot_energy_log.py`.

```
# Line of Sight (LOS)
./waf --run "nb-scenario5 --
ns3::WinnerPlusPropagationLossModel::LineOfSight=true --simTime=3600"

# Non-Line of Sight (NLOS)
./waf --run "nb-scenario5 --
ns3::WinnerPlusPropagationLossModel::LineOfSight=false --simTime=3600"

# Urban Micro Environment (UMi) vs Urban Macro (UMa)
./waf --run "nb-scenario5 --
ns3::WinnerPlusPropagationLossModel::Environment=UMi --simTime=3600"
```

Discussion:

- Compare packet delivery statistics (PCAP traces) under **LOS** vs **NLOS** environments.

Conclusion

You now have a fully functional NS-3 NB-IoT simulation environment using Docker. This setup ensures consistency across all student machines and simplifies the process of running simulations during lab sessions.